THE AGENT BROKE IT, THE AGENT FIXED IT

SELF-HEALING CODE ROLLOUTS WITH AGENTIC AI



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From 2h ago

08.47 CEST

Crowdstrike identifies issue behind worldwide outage

 Josh Taylor

Cybersecurity software firm Crowdstrike has posted in its support updates that it has identified the issue behind today's massive, global Windows outage.

It is an issue with "content deployment" that has since had those changes reverted and has advised a workaround for affected users.

We are still waiting for confirmation or a statement from Crowdstrike's media team. It's almost midnight in the US, so it's not clear when we might get more detail.

+ Show more

WHAT COULD HAVE MITIGATED THE IMPACT?

6. Template Instances should have staged deployment

Findings: Each Template Instance should be deployed in a staged rollout.

Mitigation: The Content Configuration System has been updated with additional deployment layers and acceptance checks

Staged deployment mitigates impact if a new Template Instance causes failures such as system crashes, false-positive detection volume spikes or performance issues. New Template Instances that have passed canary testing are to be successively promoted to wider deployment rings or rolled back if problems are detected. Each ring is designed to identify and mitigate potential issues before wider deployment. Promoting a Template Instance to the next successive ring is followed by additional bake-in time, where telemetry is gathered to determine the overall impact of the Template Instance on the endpoint.

PROGRESSIVE DELIVERY

Progressive Delivery is a term that includes deployment strategies that try to avoid the pitfalls of all-or-nothing deployment strategies

New versions being deployed do not replace existing versions but run in parallel for an amount of time receiving live production traffic, and are evaluated in terms of correctness and performance before the rollout is considered successful.

- Avoiding downtime
- Limit the blast radius
- Shorter time from idea to production

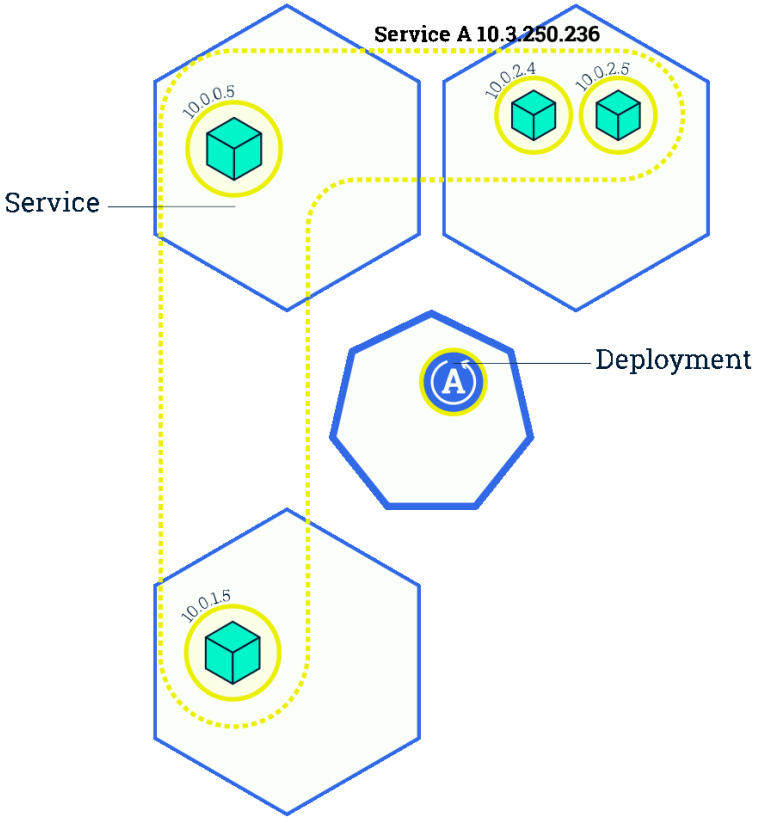
Rolling out changes to all users at once is risky

Canary rollouts and feature flags are safer

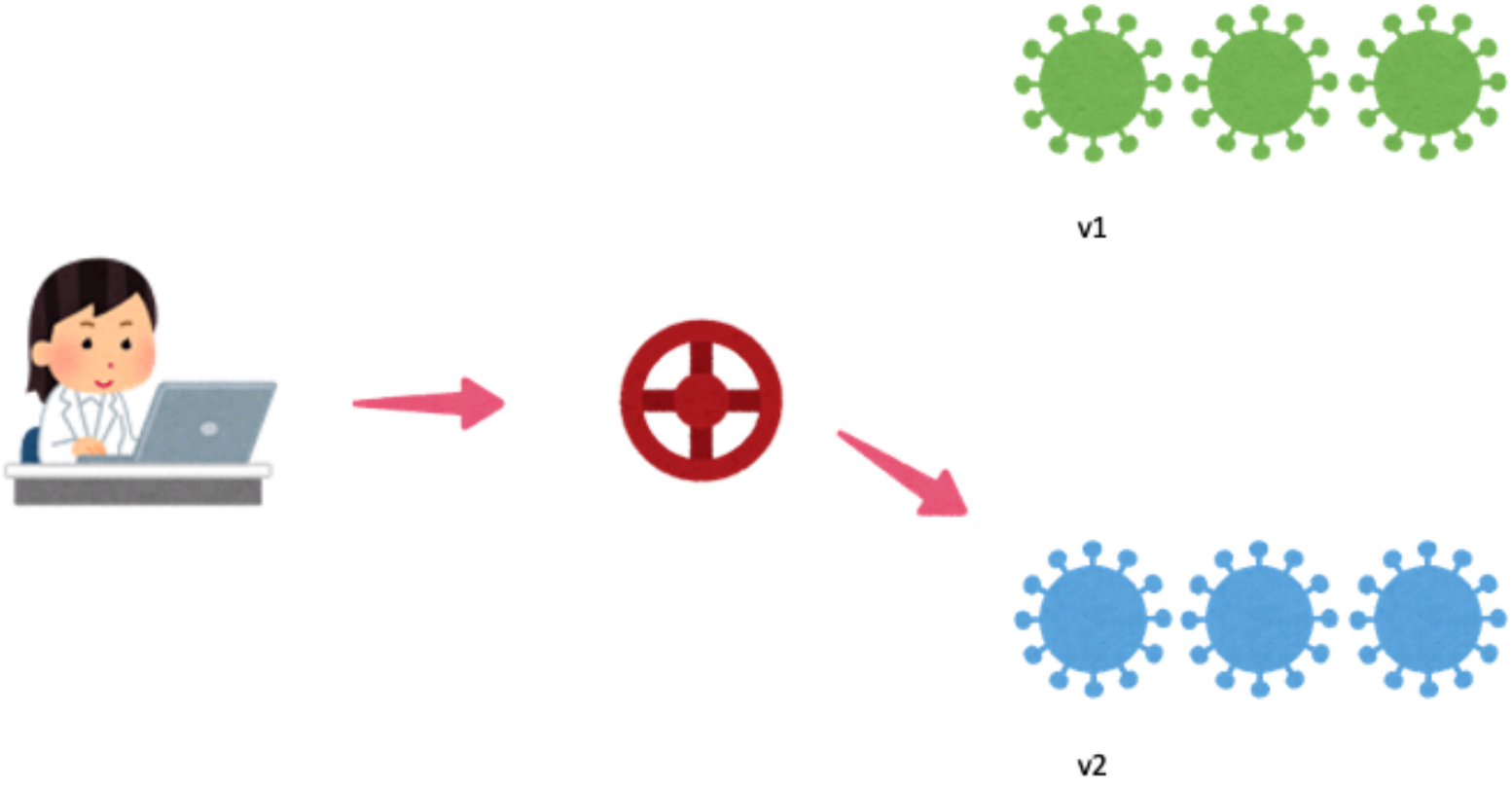
AI Agents can automate the loop fixing the failures

PROGRESSIVE DELIVERY TECHNIQUES

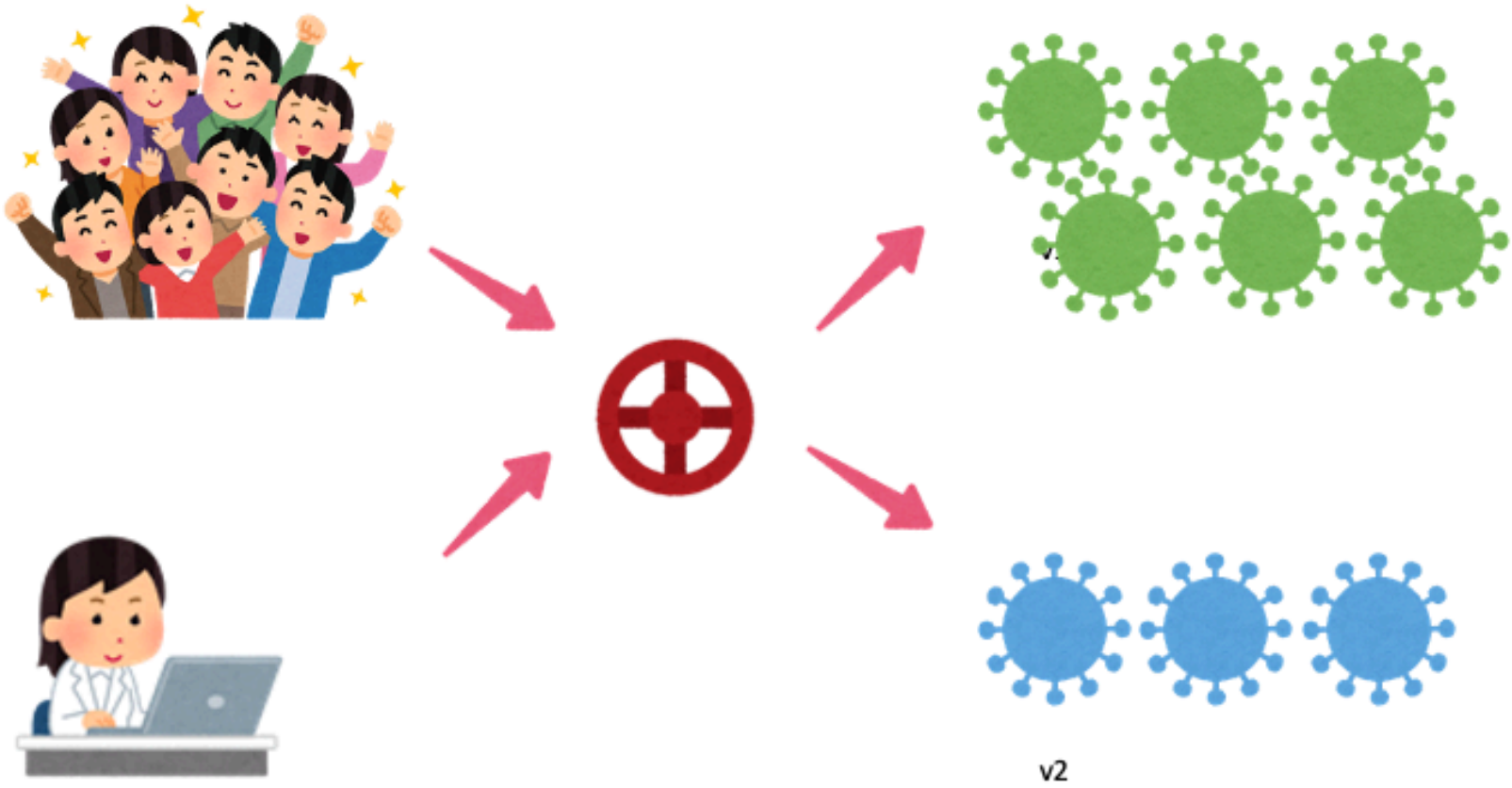
ROLLING UPDATES



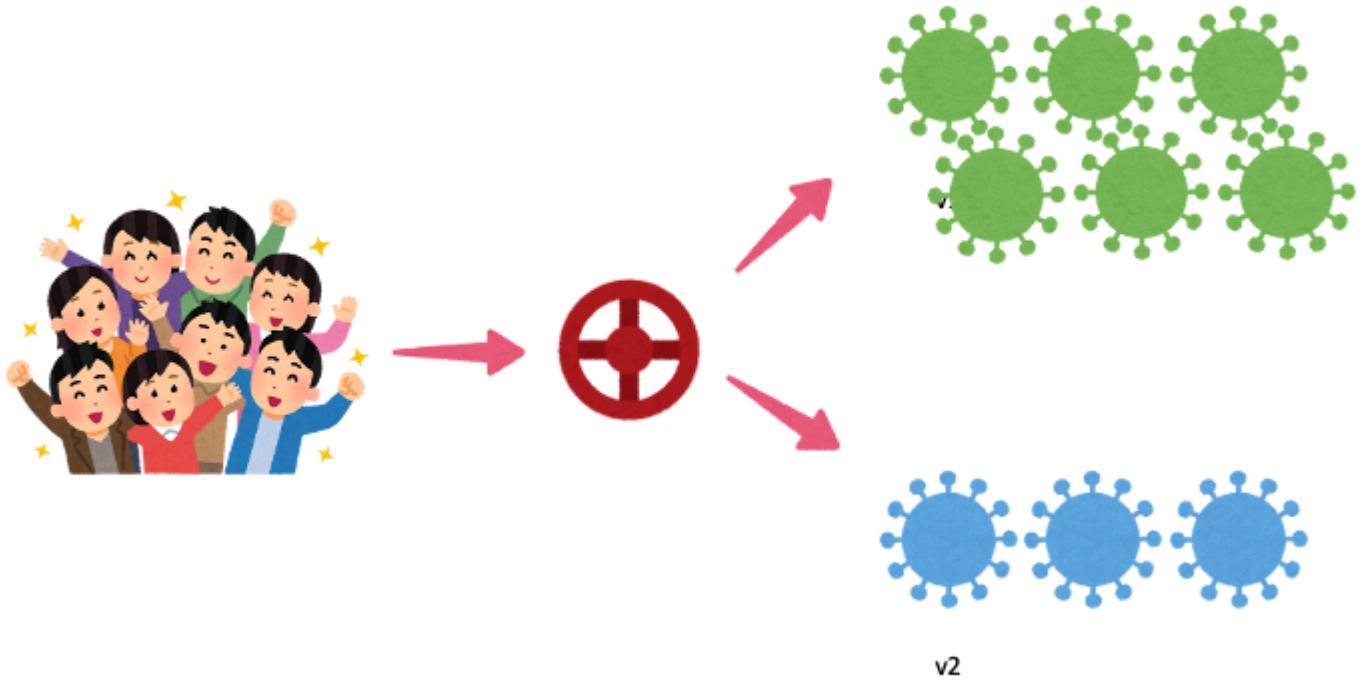
BLUE-GREEN DEPLOYMENT



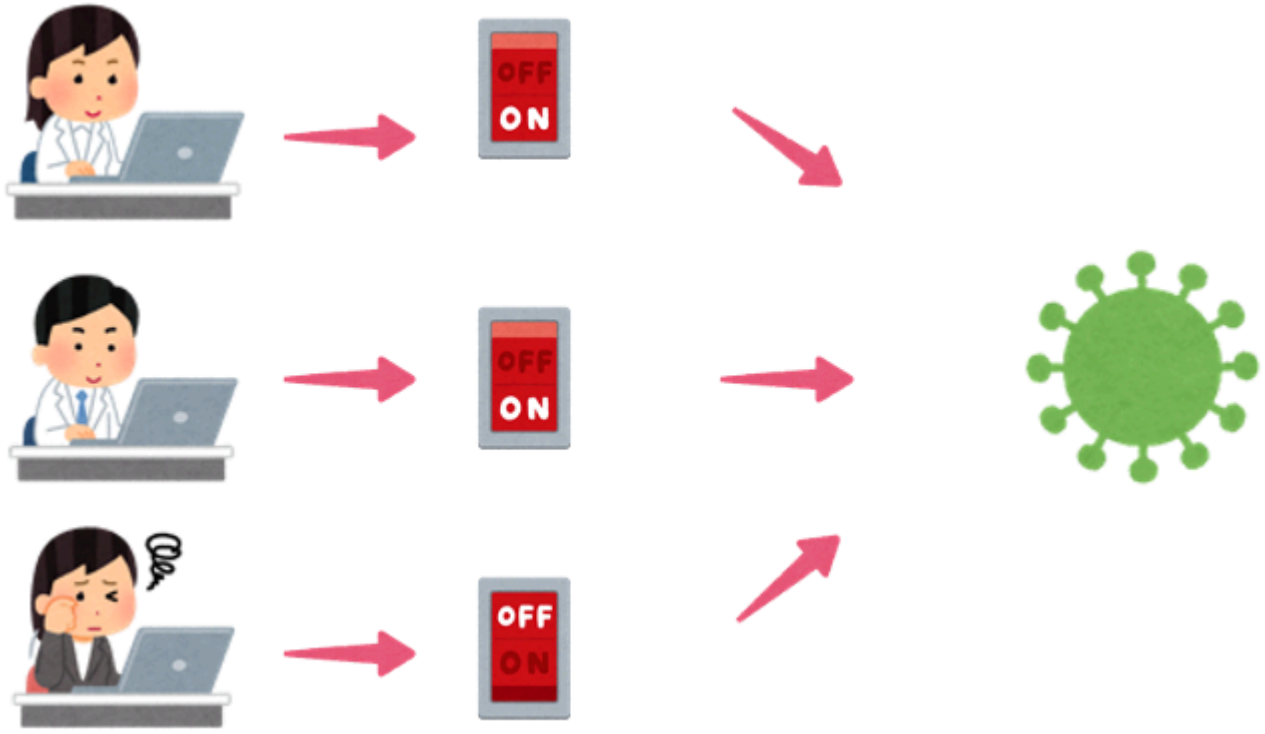
CANARY DEPLOYMENT



DARK LAUNCHING



FEATURE FLAGS



MONITORING IS THE NEW TESTING

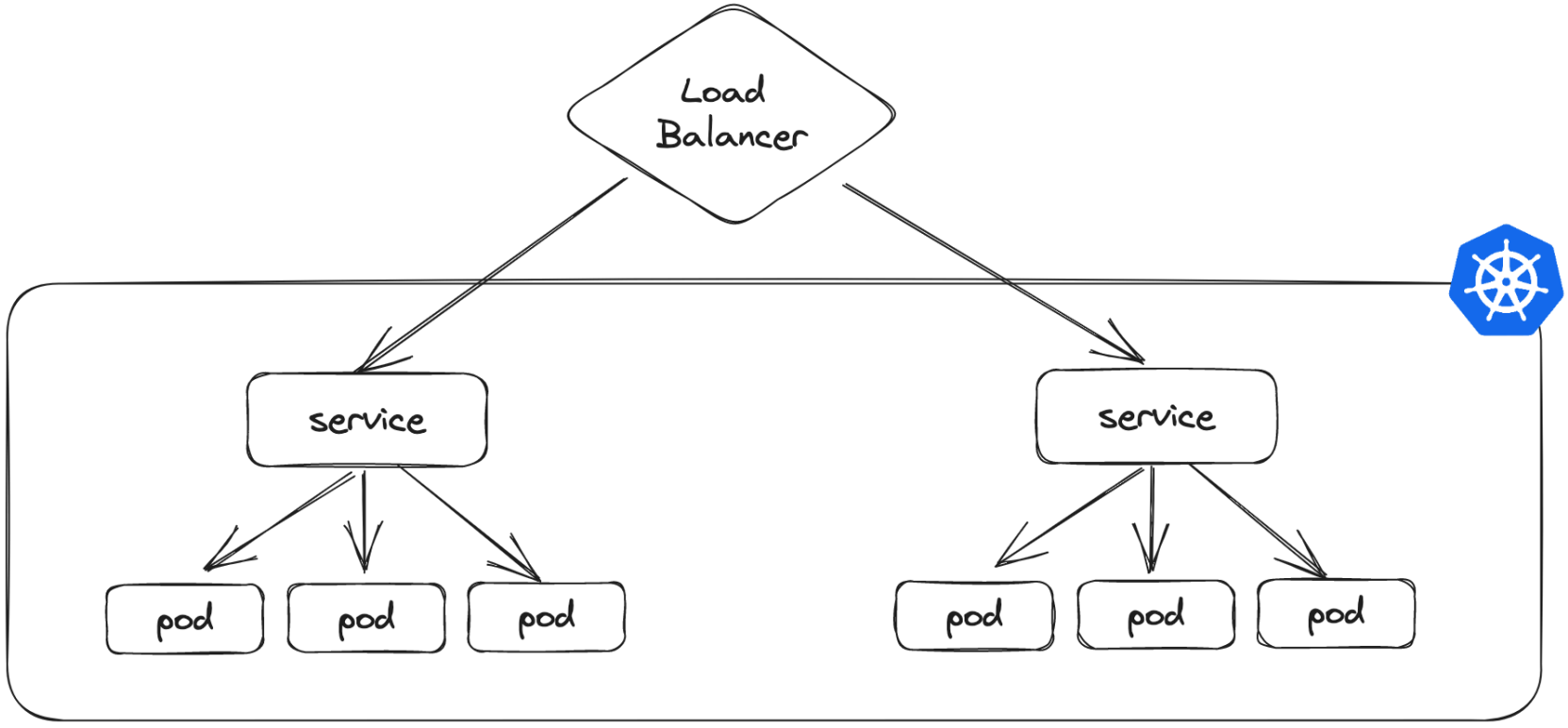
Know when users are experiencing issues **in production**

React to the issues **automatically**

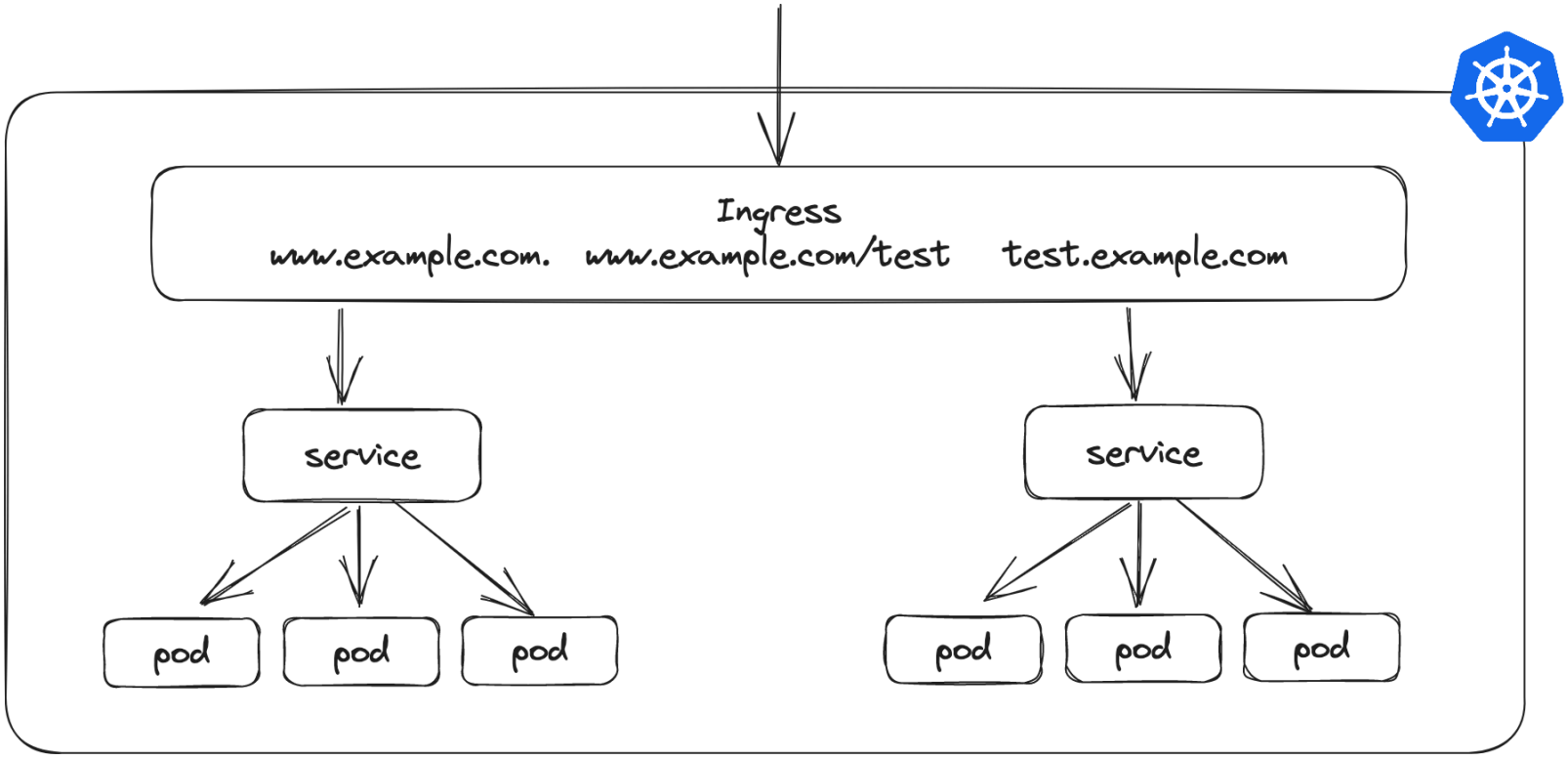
If you haven't automatically destroyed something by mistake, you are not automating enough

PROGRESSIVE DELIVERY IN KUBERNETES

KUBERNETES SERVICE ARCHITECTURE



KUBERNETES INGRESS ARCHITECTURE



KUBERNETES INGRESS

Ingress controllers:

- AWS
- GCE
- nginx
- Ambassador
- Istio Ingress
- Traefik
- HAProxy
- ...

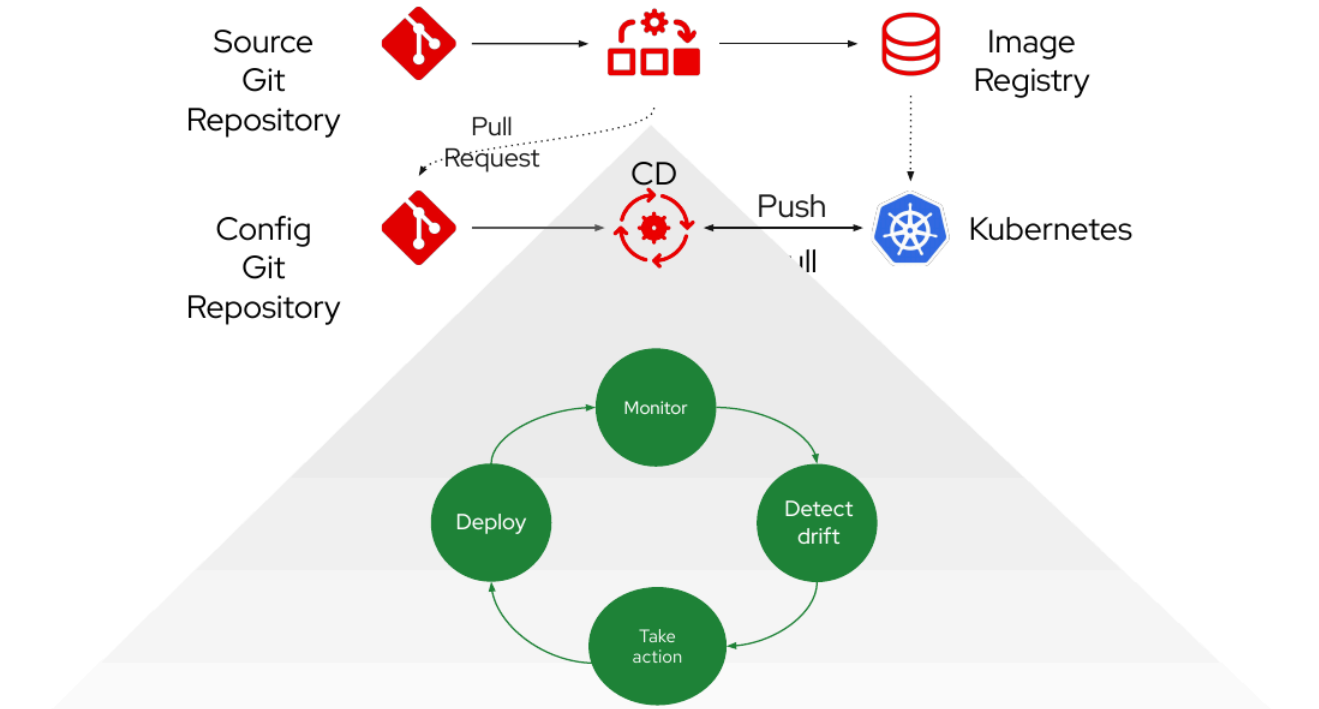
GITOPS

Git is the single source of truth

Treat everything as code

Operations through Git workflows

GitOps Application Delivery Model



BY DEFAULT:

A big bang / all or nothing release

ARGO ROLLOUTS



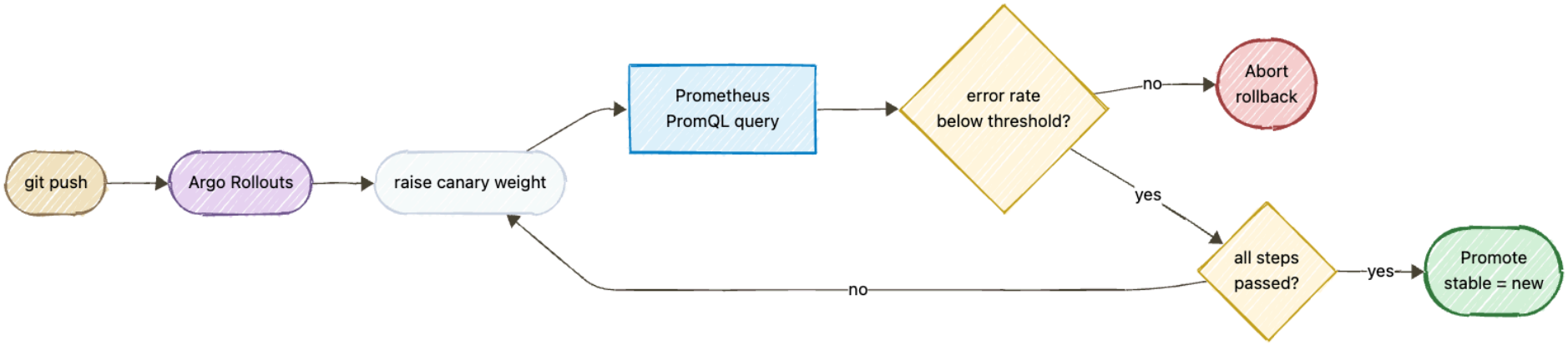
Argo Rollouts



flagger



INCREASE PERCENTAGE OF NEW (CANARY) VERSION

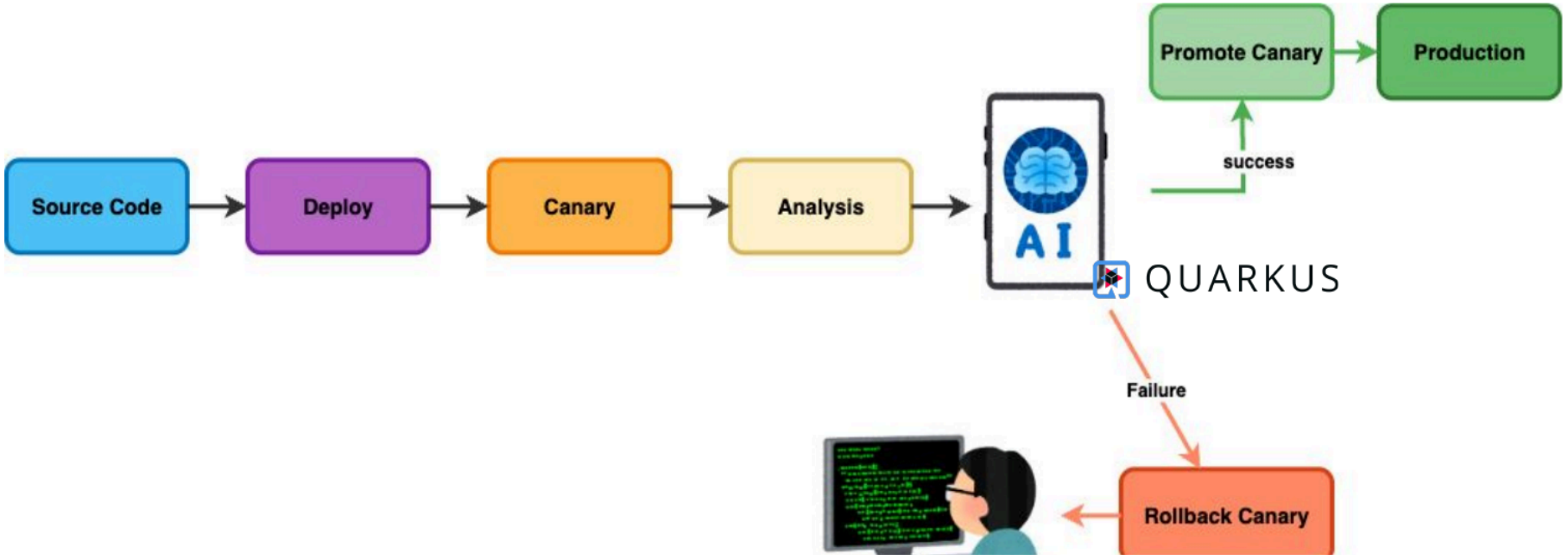


CONTINUOUS ANALYSIS WITH PROMETHEUS

```yaml
apiVersion: argoproj.io/v1alpha1
kind: AnalysisTemplate
metadata:
  name: success-rate
spec:
  args:
  - name: service-name
  metrics:
  - name: success-rate
    interval: 10s
    successCondition: len(result) == 0 || result[0] >= 0.95
    failureLimit: 2
    provider:
      prometheus:
        address: http://prometheus.istio-system.svc.cluster.local:9090
        query: |
          sum(irate(istio_requests_total{
            reporter="source",
            destination_service=~"{{args.service-name}}",
            response_code!~"5.*"}[30s])
          )
```

AI-POWERED ROLLOUT ANALYSIS


Write
custom
PromQL
Use AI
to analyze
failures
imgflip.com


Source Code
Deploy
Canary
Analysis
AI
QUARKUS
Promote Canary
Production
success
Failure
Rollback Canary

Argo Rollouts Canary Analysis with AI Plugin

Configuration

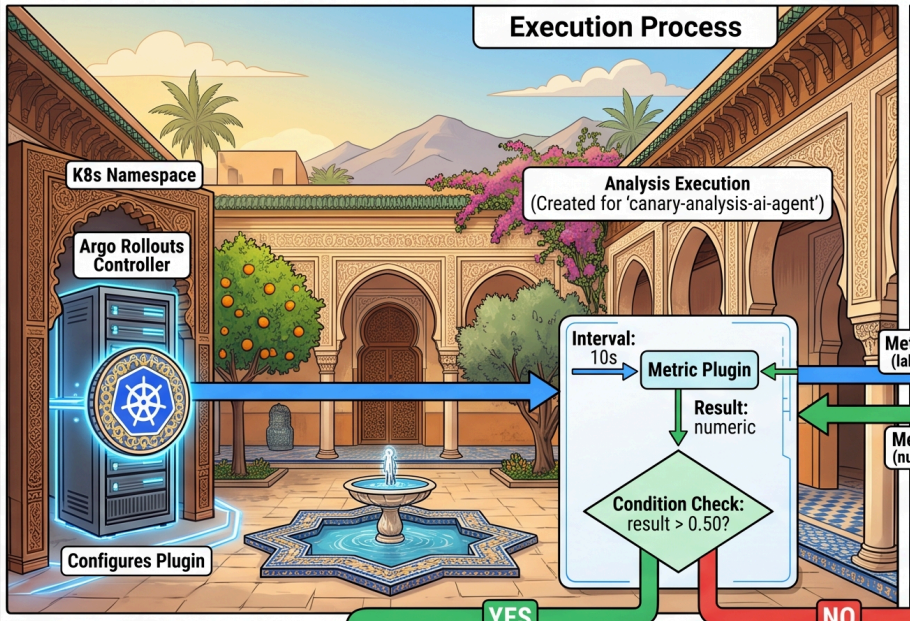
AnalysisTemplate
(canary-analysis-ai-agent)
Metrics:
- success-rate
plugin: argoproj-labs/metric-ai

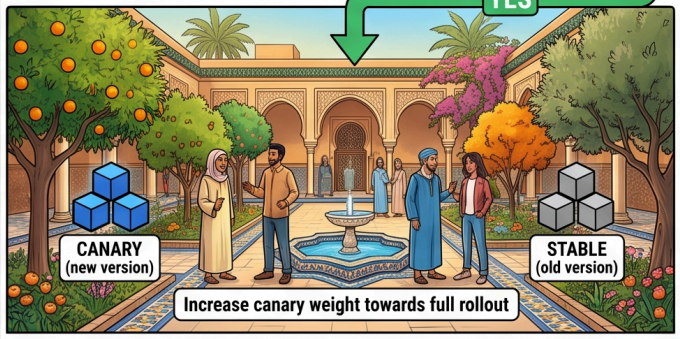
Success Condition: result > 0.50

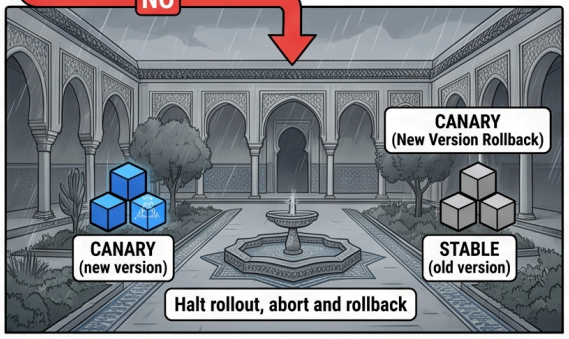
Rollout:
Strategy: Canary
analysis:
templates [canary-analysis-ai-agent]
metadata:
canary: role=canary
stable: role=stable

RolloutManager:
Manager
plugins:
metric: argoproj-labs/metric-ai

Execution Process







REGISTER THE AI METRIC PLUGIN

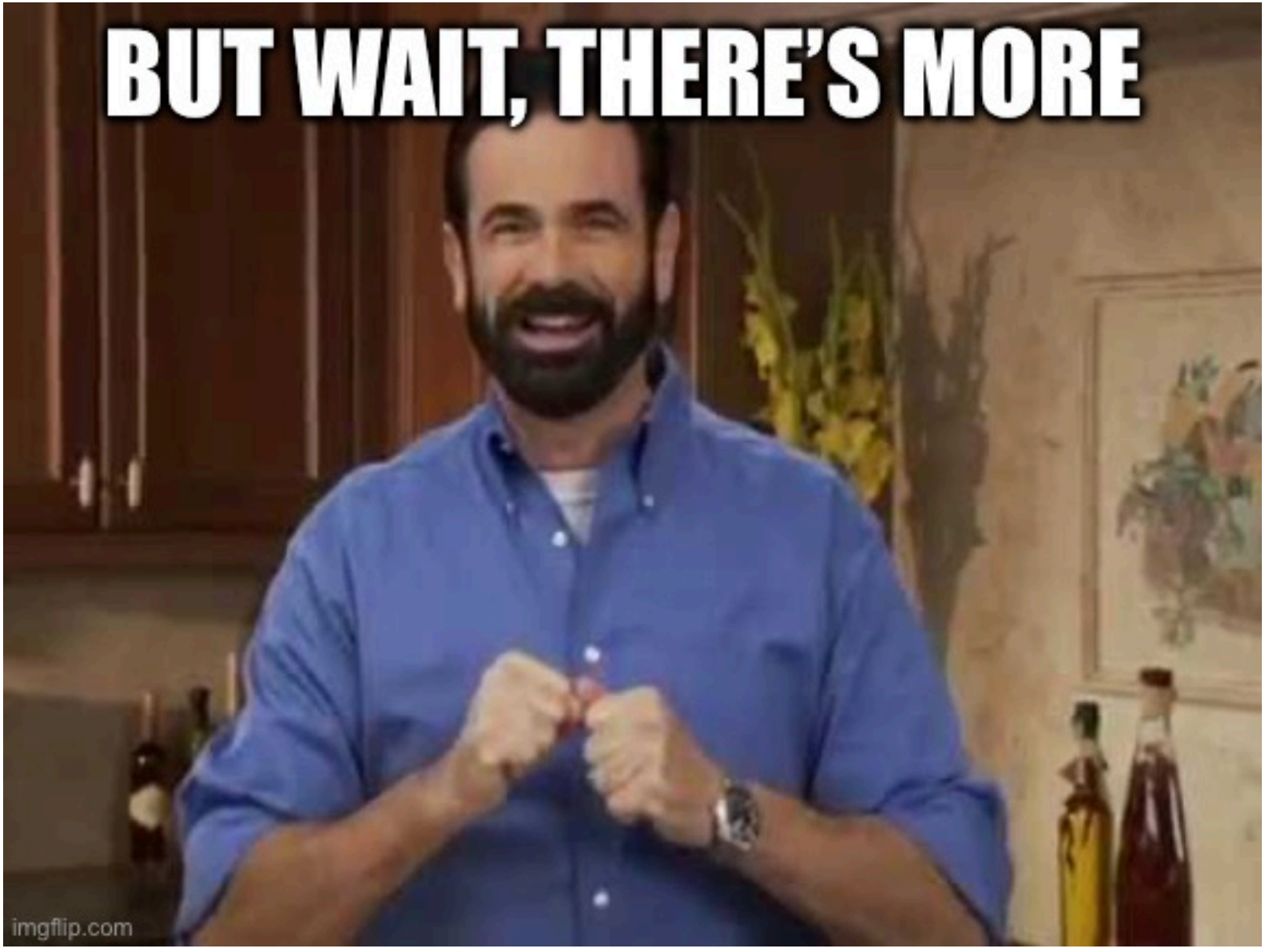
```
apiVersion: argoproj.io/v1alpha1
kind: RolloutManager
metadata:
  name: argo-rollouts
spec:
  plugins:
    metric:
    - name: argoproj-labs/metric-ai
      location: >-
        https://github.com/argoproj-labs/rollouts-plugin-metric-
ai/releases/download/v0.0.1/rollouts-plugin-metric-ai-linux-amd64
```

CONFIGURE CANARY ANALYSIS

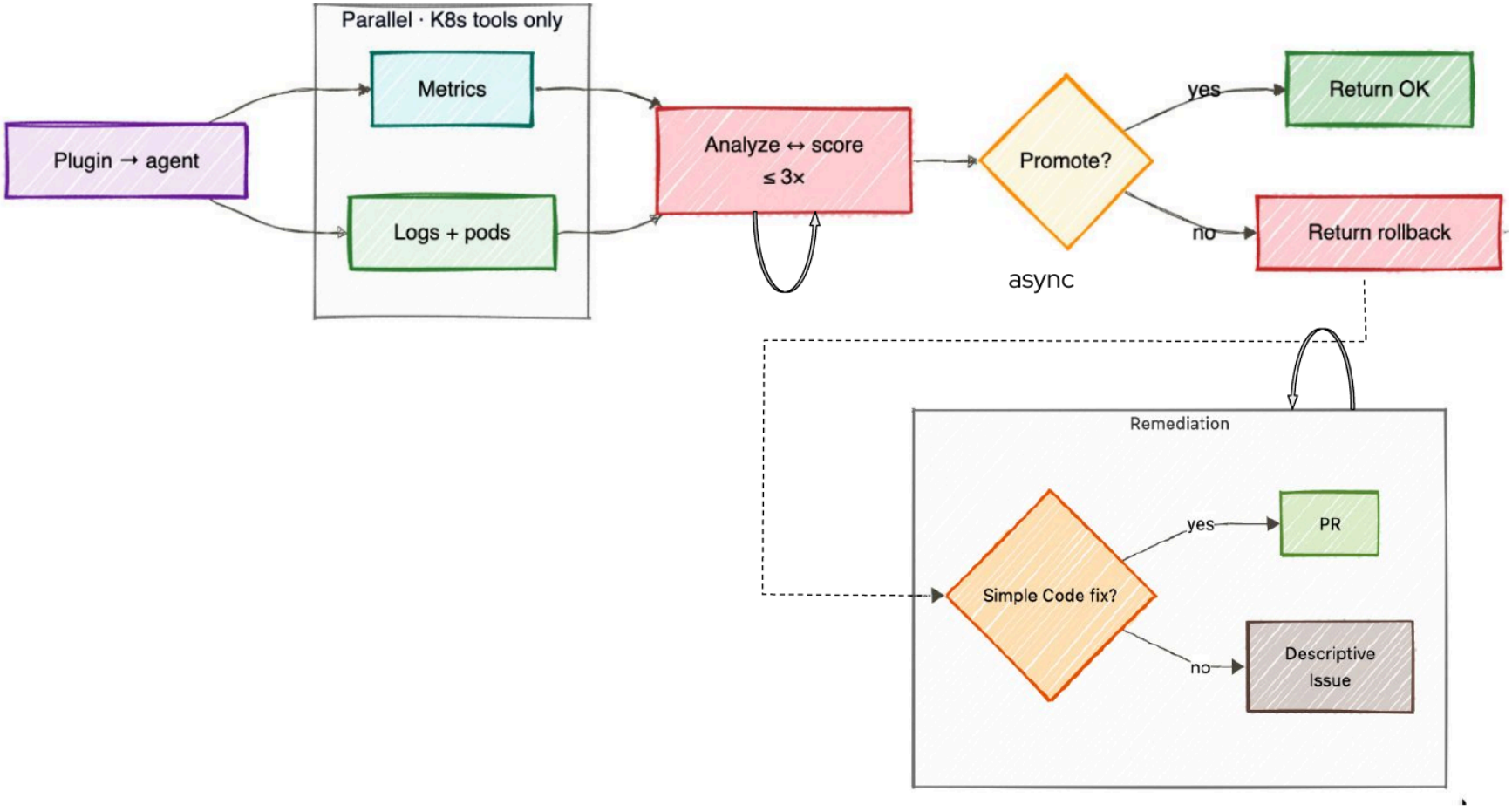
```yaml
canary:
  analysis:
    startingStep: 1
    templates:
    - templateName: canary-analysis-ai-agent
  canaryMetadata:
    labels:
      role: canary
  stableMetadata:
    labels:
      role: stable
```

LET THE AI AGENT EVALUATE THE ROLLOUT

```yaml
apiVersion: argoproj.io/v1alpha1
kind: AnalysisTemplate
metadata:
  name: canary-analysis-ai-agent
spec:
  metrics:
  - interval: 10s
    name: success-rate
    provider:
      plugin:
        argoproj-labs/metric-ai:
          agentUrl: http://kubernetes-agent:8080
          stableLabel: role=stable
          canaryLabel: role=canary
          extraPrompt: ignore aesthetic changes
    successCondition: result > 0.50
```


BUT WAIT, THERE'S MORE
imgflip.com

ASYNCHRONOUS REMEDIATION



GIVE THE AGENT THE SOURCE REPOSITORY

```yaml
apiVersion: argoproj.io/v1alpha1
kind: AnalysisTemplate
metadata:
  name: canary-analysis-ai-agent
spec:
  metrics:
  - interval: 10s
    name: success-rate
    provider:
      plugin:
        argoproj-labs/metric-ai:
          agentUrl: http://kubernetes-agent:8080
          stableLabel: role=stable
          canaryLabel: role=canary
          extraPrompt: ignore aesthetic changes
          githubUrl: https://github.com/kdubois/argo-rollouts-quarkus-demo
```

DEMO



LESSONS LEARNED

Performance was an interesting challenge

From one AI service to an agentic system: parallel and async agents

LLM CHOICE + CONTEXT ENGINEERING + TOOL CALLING

Especially for PR creation

COMPLEXITY VS PORTABILITY

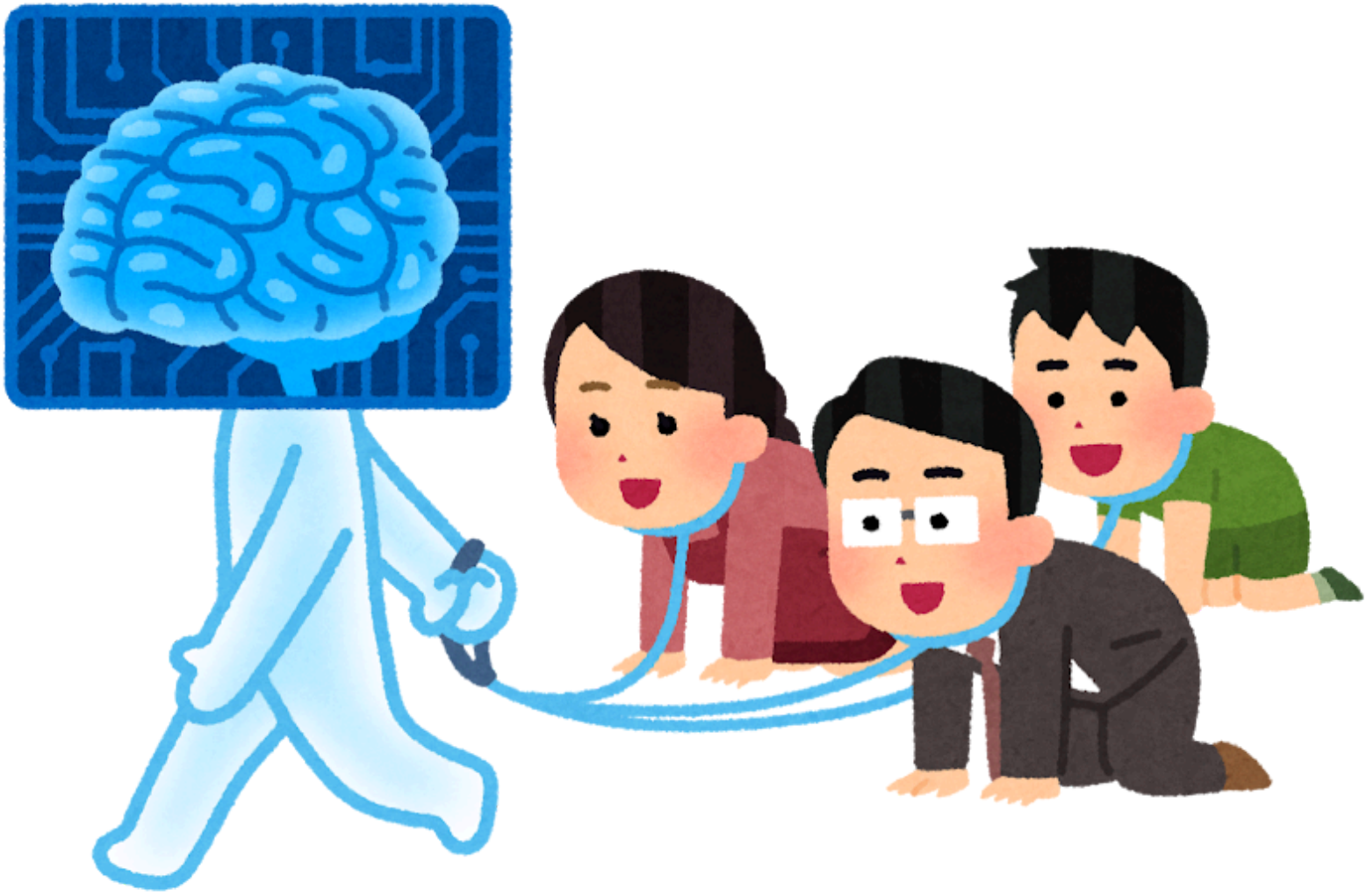
Could have used Serverless MCP, an external code assistant for PR creation, async remote agents, etc.

TAKEAWAYS

Rolling out changes to all users at once is risky

Canary rollouts and feature flags are safer

AI agents can automate the loop by analyzing metrics and logs, and even proposing fixes for failures



RESOURCES

Argo Rollouts Plugin for AI metrics

rollouts-plugin-metric-ai

Demo app

argo-rollouts-quarkus-demo

Antigravity SDK Agents

kubernetes-agents-antigravity

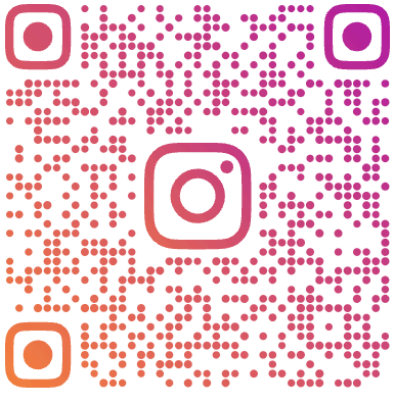
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